CAMARINES SUR POLYTECHNIC COLLEGES

COLLEGE OF COMPUTER STUDIES

CCCS 106: Application Development and Emerging Technologies (AY 2026-2027)

LABORATORY TASK REPORT 01

Environment Isolation, Modern Flet GUI & Collaborative 3-Developer Git Workflow

Group Designation:	Group No. 12
Date of Submission:	September 9, 2026
Course Instructor:	Engr. Allan Ibo Jr. | CSPC College of Computer Studies

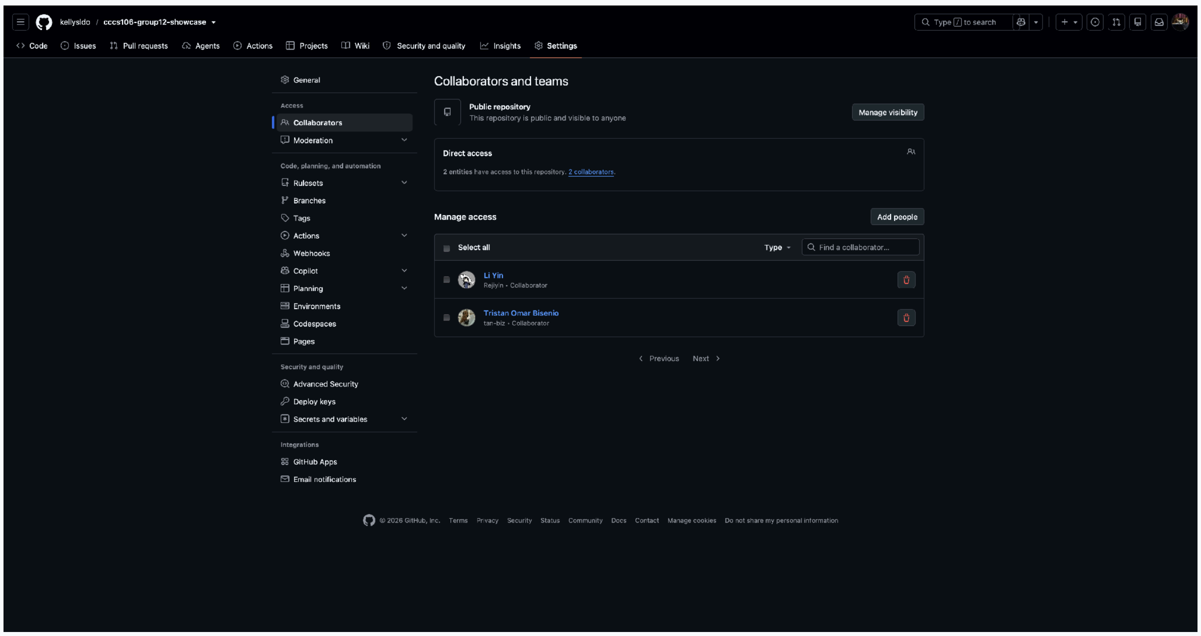
TEAM TRIAD MEMBERS & TECHNICAL ROLES

Student Full Name	Student ID	GitHub Username	Assigned Triad Role & Branch
Kelly Saldo	2411287	@kellysldo	Developer 1 (Lead UI) feature/dev1-profile
Tristan Bisenio	232000006	@tan-biz	Developer 2 (State Logic) feature/dev2-state
Rizelyn Joy Borbe & Nash Sabas	2410605 2412117	@rejiyin @Nash14sabas	Developer 3 (QA & Milestone) feature/dev3-features

Section 1: GitHub Repository & Collaborator Proof

Public GitHub Repository URL: https://github.com/kellysldo/cccs106-group12-showcase

Proof of team collaboration setup on GitHub. The screenshot below shows the repository Settings > Collaborators page confirming that Developer 1 has invited Developer 2 and Developer 3 as official contributors with accepted statuses.



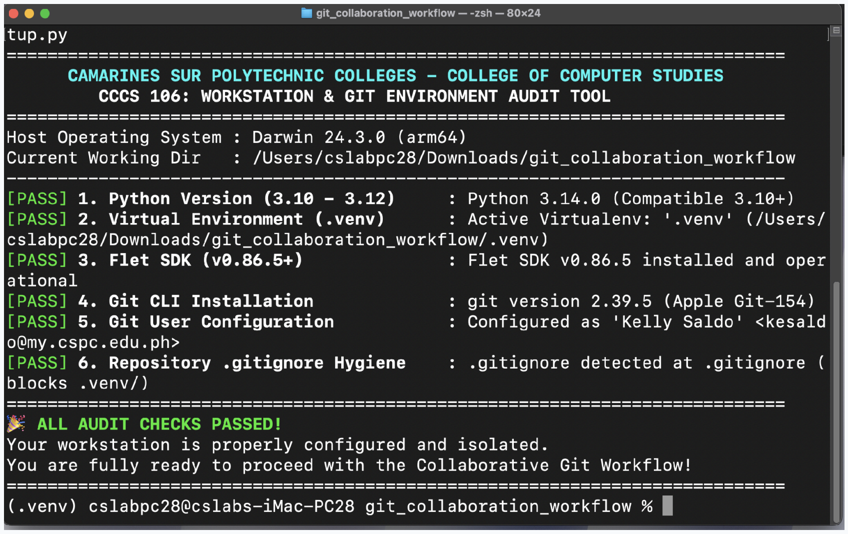
Reference: GitHub Collaborators Verification (Settings > Collaborators)

Requirement: Screenshot showing Developer 1 (Owner), Developer 2, and Developer 3 with accepted invitations.

Figure 1: GitHub Collaborators Verification (Settings > Collaborators)

Section 2: Automated Pre-Flight Diagnostics Audit

Pre-flight workstation audit executed via 'python verify_setup.py' inside the active virtual environment (.venv). All six (6) audit checks must report [PASS] verifying Python 3.10+, virtualenv isolation, Flet SDK v0.86.5+, Git binary detection, configured institutional user identity, and .gitignore hygiene.



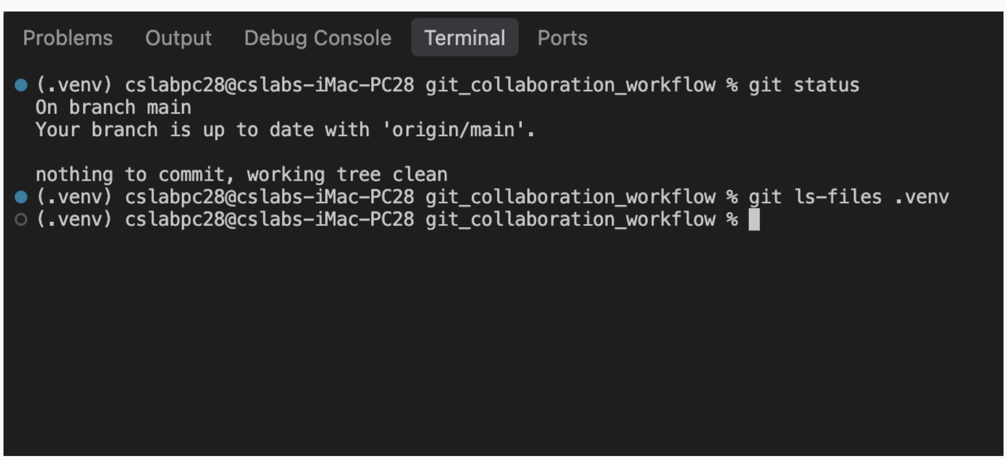
Reference: Pre-flight Environment Diagnostics Audit (verify_setup.py)

Requirement: Terminal screenshot displaying green CSPC header and 6/6 [PASS] checks.

Figure 2: Pre-flight Environment Diagnostics Audit (verify_setup.py)

Section 3: Git Working Tree Cleanliness & .gitignore Audit

Demonstration of repository hygiene. The screenshot proves that the working tree is clean with zero uncommitted changes, and that .venv/ is completely untracked (verified by running 'git ls-files .venv', returning empty output).



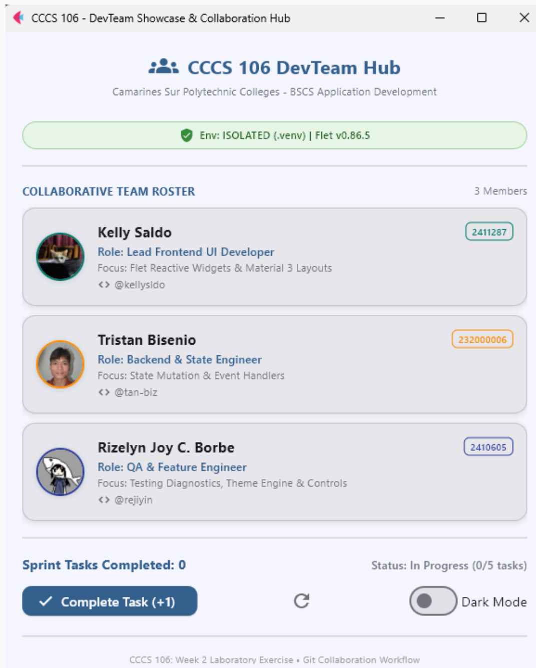
Reference: Git Working Tree Cleanliness & .gitignore Hygiene Proof

Requirement: Terminal screenshot of 'git status' reporting working tree clean and empty 'git ls-files .venv'.

Figure 3: Git Working Tree Cleanliness & .gitignore Hygiene Proof

Section 4: Running Flet v0.86.5 Application Showcase

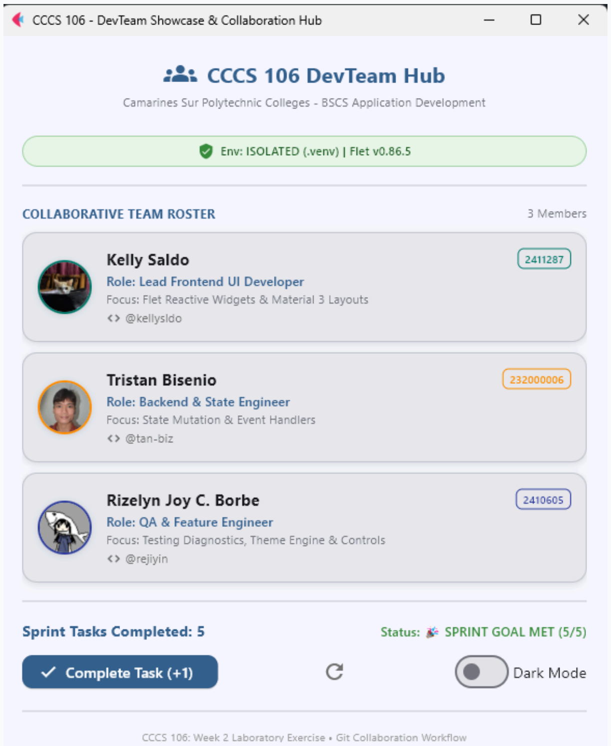
Desktop window screenshots demonstrating the running collaborative Flet GUI application across its three core interactive states: Initial Launch, Sprint Goal Milestone Reached, and Dark Mode theme toggle.



Reference: Application Initial Launch State

Requirement: Window showing CSPC header, green ISOLATED (.venv) badge, 3 team member cards, 0 tasks completed, and In Progress status.

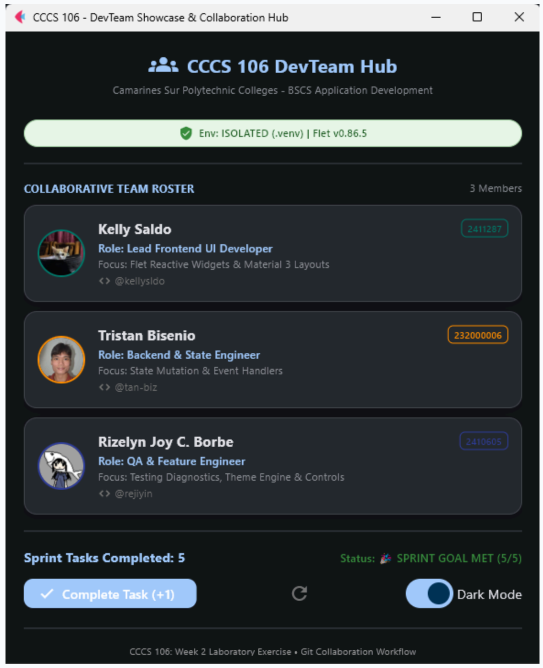
Figure 4: Application Initial Launch State



Reference: Sprint Goal Milestone Reached State

Requirement: Window showing counter at 5, status badge 'SPRINT GOAL MET (5/5)' in green, and Complete Task button disabled.

Figure 5: Sprint Goal Milestone Reached State



Reference: Material 3 Dark Mode Theme Toggle State

Requirement: Window with Dark Mode toggle active, demonstrating reactive Material 3 dark surface container rendering.

Figure 6: Material 3 Dark Mode Theme Toggle State

Section 5: Triad Contribution Matrix & Learning Reflections

Summary of individual contributions and commit accountability across all three triad developers.

Member Name	Role	Branch Name	Commit SHA	Specific Engineering Contributions
Kelly A. Saldo	Developer 1 (Lead UI)	feature/dev1-profile	[fa4170d, 371472b, b2ae4cf, 5de01c8, e012785, c9c06be, 727d02d, 67583d8, eae0784, b467486, b467486]	Initialized repo, setup .gitignore, integrated card 1, configured remote repo & pushed main.
Tristan Bisenio	Developer 2 (State Logic)	feature/dev2-state	[cd4c84e, 7d0cc98, 7c13cf0, 3ee6ab7, 4f143c6, dc073b0, 9988ac1]	Synced main, integrated card 2, connected increment/reset event handlers, merged & pushed.
Rizelyn Joy C. Borbe	Developer 3 (QA/Feature)	feature/dev3-features	[0c25ac1, 14d0d91, 1fe62ff, b9a76f0]	Synced main, integrated card 3, built milestone badge logic & boundary constraints, merged & pushed.

Individual Technical Reflections

Developer 1 (Lead UI & Repo Architect):

Through this learning task, I learned that using a virtual environment isolates a project's Python packages and dependencies, helping to avoid conflicts with other projects on the same computer. I also learned how .gitignore must be configured carefully so that unnecessary or sensitive data files like virtual environments are excluded and while keeping important project files tracked. I also learned while working in GitHub, coordinating repository invitations needs to ensure that team members receive access and accept invitations before collaborating. Overall, I learned with the proper environment isolation, Git configuration, and team coordination and collaboration really helps to keep the project organized, easy to maintain, secured, and how GitHub can cater all the necessary needs for a team collaboration on a project.

Developer 2 (State & Event Logic Engineer):

Through this task, I learned that cloning a remote repository lets each developer get their own copy of the project on their computer while still staying connected to the shared GitHub repo. I also learned why it's important to run git pull origin main before starting new work. It keeps my local files up to date and helps prevent conflicts when other members are also making changes. While building the increment and reset event handlers, I saw how Flet updates the screen automatically whenever the state changes, so I didn't have to manually refresh anything. Overall, this exercise showed me that cloning, syncing, and writing clear event handlers all help the team work together smoothly without breaking each other's code.

Developer 3 (QA & Sprint Goal Feature Engineer):

[Rizelyn Joy Borbe] Working on feature/dev3-features gave me a real appreciation for feature branching while building the sprint goal milestone logic and boundary limits. Since my task involved testing edge cases—like making sure the task counter capped at 5 and properly triggered the "SPRINT GOAL MET" badge—having an isolated branch meant I could test state limits and debug UI behavior freely. It kept main clean while Developer 1 and Developer 2 pushed their profile and state changes. This hands-on experience showed me that branching isn't just a requirement; it's what lets us test edge cases and ship feature logic without risking the overall application stability.

[Nash Sabas] This hands-on exercise highlighted the importance of using virtual environments, it can provide a project with specific packages whilst not interfering with the global installation of Python. Working as the QA and Sprint Goal Feature Engineer taught me the necessity of isolating feature branches in order to maintain safety and stability while code updates are shipped. Overall, I learned to enforce stringent constraints across state borders so user data and UI states do not leak into other unrelated screens during times when tests need to be performed rapidly. It has been very important that I have been able to identify issues with Flet’s Material 3 controls wherein theme inconsistencies and issues with layout scaling have appeared in the course of testing.

Appendix: Verbatim Git Commit Tree Log

Verbatim ASCII commit history exported using 'git log --oneline --graph --all > git_log_proof.txt'. This demonstrates the complete sequential feature branching and merge timeline across all three developers.

```
*   fa4170d Merge pull request #5 from kellysldo/feature/dev1-profile
|\
| * 371472b docs(screenshots): add laboratory verification proofs and feat(profile):
integrate developer 1 profile card
| * b2ae4cf feat(profile): integrate developer 1 profile card
* |   9988ac1 Merge pull request #4 from kellysldo/feature/dev2-state
|\ \
| * | dc073b0 Add files via upload
* | | 979a0ce Delete Downloads/cccs106-group12-showcase-main
(1)/cccs106-group12-showcase-main/screenshots directory
* | | 36ee70d Delete Desktop/cccs106-group12-showcase-main/screenshots directory
| |/
|/|
* |   c533926 Merge branch 'main' of https://github.com/kellysldo/cccs106-group12-showcase
|\ \
| * \   4f143c6 Merge pull request #3 from kellysldo/feature/dev2-state
| |\ \
| | |/
| | * 3ee6ab7 Update team_profiles.py
| * |   7c13cf0 Merge pull request #2 from kellysldo/feature/dev2-state
| |\ \
| | |/
| | * 7d0cc98 Update team_profiles.py
| * | 727d02d Delete assets/dev1.png
| * | c9c06be Add files via upload
| * | 0c25ac1 feat(milestone): integrate developer 3 profile and sprint goal milestone
badge
| |/
| *   e15e295 Merge pull request #1 from kellysldo/feature/dev2-state
| |\
| | * cd4c84e feat(state): integrate developer 2 profile and task counter handlers
| * | 07ad32f docs(screenshots): add laboratory verification proofs
| * | 67583d8 feat(profile): integrate developer 1 profile card
| * | eae0784 feat(profile): integrate developer 1 profile card
| |/
| * b467486 chore(init): baseline repository with modern flet app and verification tool
* 5de01c8 docs(screenshots): add laboratory verification proofs
* e012785 docs(screenshots): add laboratory verification proofs
```